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**A mobile-integrated deep learning system for rapid screening
of respiratory diseases using cough sound analysis**

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**DATE
23 – APRIL - 2026**

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LIST OF ABBREVIATIONS

AI — Artificial Intelligence

AUC — Area Under the Receiver Operating Characteristic Curve

CNN — Convolutional Neural Network

COVID-19 — Coronavirus Disease 2019

CSV — Comma-Separated Values

dB — Decibel

DL — Deep Learning

FFT — Fast Fourier Transform

FNR — False Negative Rate

FPR — False Positive Rate

JSON — JavaScript Object Notation

MFCC — Mel-Frequency Cepstral Coefficients

ML — Machine Learning

ROC — Receiver Operating Characteristic

SNR — Signal-to-Noise Ratio

STFT — Short-Time Fourier Transform

TB — Tuberculosis

TFLite — TensorFlow Lite

WHO — World Health Organization

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Respiratory diseases continue to impose a major burden on public health worldwide and remain among the leading causes of death and disability, especially in low-resource settings where access to diagnosis and treatment may be delayed (WHO, 2023). In many underserved environments, screening and diagnosis are limited by the high cost of confirmatory tests, shortages of trained personnel, and restricted access to healthcare facilities (WHO, 2023). Such delays may worsen disease progression and, in the case of infectious respiratory illnesses, may contribute to ongoing transmission within communities.

At the same time, smartphone adoption has increased significantly, making microphones, storage, and moderate computational resources widely available to ordinary users. This development creates an opportunity to design mobile health tools capable of supporting early screening outside conventional healthcare settings. A smartphone-based screening tool does not replace clinical diagnosis, but it can serve as an accessible and low-cost triage mechanism that identifies potentially high-risk cases and encourages timely confirmatory testing (Imran et al., 2020; Schuller et al., 2021).

Cough is one of the most common symptoms associated with respiratory disease. The acoustic characteristics of cough sounds are influenced by airway dynamics, inflammation, mucus presence, and other physiological changes. Although clinicians can distinguish certain coughs in broad descriptive terms such as dry or productive, subtle disease-related patterns are often beyond reliable human auditory interpretation. Machine learning and deep learning methods can learn such patterns from large labeled datasets and offer an objective route to automated analysis (Laguarta et al., 2020; Brown et al., 2020).

Research in cough-based screening expanded rapidly during and after the COVID-19 pandemic. Several public datasets now provide respiratory audio and metadata at a scale sufficient for model development. For example, Coswara provides breathing, cough, and voice recordings for COVID-19-related analysis, while COUGHVID offers a large crowdsourced cough corpus for algorithm development (Sharma et al., 2020; Orlandic et al., 2021). However, moving from proof-of-concept models to practical smartphone-based systems requires more than model training. It requires robust data governance, explicit label mapping, careful split design to reduce leakage, efficient preprocessing, compact feature extraction, reproducible training procedures, and mobile-oriented deployment pathways (Kapetanidis et al., 2024; Schuller et al., 2021).

This dissertation is therefore motivated by the need to move from concept-level respiratory sound classification toward a documented, reproducible, and deployment-oriented cough-screening pipeline. While the broader long-term aim includes multi-class respiratory disease screening, the current implemented milestone focuses on a binary screening task using publicly accessible COVID-related cough datasets, thereby establishing the foundation for later extension to additional diseases such as tuberculosis.

1.2 Problem Statement

Traditional screening and diagnostic methods for respiratory diseases often require clinical facilities, specialized tests, and trained personnel. These conditions are not consistently available in many settings where early detection is most needed, leading to delays in care and referral (WHO, 2023).

Although numerous studies report promising results for cough-based disease classification, several practical and methodological challenges remain unresolved. Public datasets frequently contain inconsistent labels, incomplete metadata, and heterogeneous recording conditions (Orlandic et al., 2021; Sharma et al., 2020). In some studies, evaluation credibility is weakened by possible subject leakage, class imbalance, or limited reproducibility of preprocessing and split generation (Kapetanidis et al., 2024). Furthermore, many models are designed and validated under controlled conditions without

adequate consideration of mobile deployment constraints such as inference latency, compact model size, offline operation, and privacy preservation (Imran et al., 2020; TensorFlow Lite Documentation, n.d.).

The central problem addressed in this dissertation is therefore the lack of a well-documented, reproducible, and mobile-oriented end-to-end system that transforms raw cough datasets into a lightweight screening model with transparent evaluation and a deployment-ready workflow.

1.3 Rationale of the Study

This study is justified by both public health needs and technological feasibility. First, earlier screening of respiratory diseases can support faster referral, earlier intervention, and more efficient use of limited healthcare resources (WHO, 2023). Second, smartphones already provide a widely distributed platform on which screening tools can be deployed at relatively low cost (Imran et al., 2020). Third, recent advances in deep learning make it possible to learn time-frequency patterns from cough audio that may not be obvious to human listeners (Laguarta et al., 2020; Brown et al., 2020).

In addition, a major gap exists between published algorithmic results and practical implementation. Many prior studies emphasize predictive performance while under-reporting data cleaning procedures, label mapping, and deployment readiness (Kapetanidis et al., 2024). This dissertation contributes by addressing not only classification but also the full supporting pipeline: data acquisition, metadata auditing, label governance, preprocessing, feature extraction, baseline modeling, screening-oriented evaluation, and mobile export preparation.

The study is therefore significant not only for its potential health application, but also for its methodological contribution to reproducible, transparent, and deployment-conscious respiratory sound analysis.

1.4 AIM AND OBJECTIVES

1.4.1 Aim

The aim of this dissertation is to develop a mobile-integrated deep learning system for rapid screening of respiratory diseases using cough sound analysis.

1.4.2 Specific Objectives

To achieve this aim, the study pursues the following objectives:

O1. Acquire and curate cough recordings from public datasets, audit metadata quality, and create reproducible data splits.

O2. Define and implement an explicit label schema with traceable mapping rules, documented exclusions, and support for later extension to multi-class respiratory screening.

O3. Implement preprocessing and feature extraction procedures using compact log-mel spectrogram representations suitable for later mobile inference.

O4. Train a lightweight convolutional neural network baseline model appropriate for smartphone-oriented screening.

O5. Establish an evaluation framework that reports class-specific metrics, confusion matrices, and error analysis outputs for screening tasks.

O6. Prepare a deployment pathway toward mobile inference through TensorFlow Lite export and associated configuration artifacts.

At the current stage of implementation, Objectives O1 through O5 have been substantially realized for a binary COVID-versus-healthy/non-target screening task. Objective O6 has been pursued through export preparation and multiple TFLite conversion attempts, although final conversion was not stable in the current software environment.

Table 1.1: Objectives and Indicators

Objective	Main Output	Example Indicator
O1. Acquire and curate cough recordings from public datasets, audit metadata quality, and create reproducible data splits	unified metadata and splits	Cleaned metadata saved; train/validation/test CSVs generated and reusable
O2. Define and implement an explicit label schema with traceable mapping rules, documented exclusions, and support for later extension to multi-class respiratory screening	explicit label mapping functions	Raw labels mapped consistently into training labels with inclusion/exclusion notes
O3. Implement preprocessing and feature extraction procedures using compact log-mel spectrogram representations suitable for later mobile inference	cached feature arrays	Fixed-size feature arrays generated for train, validation, and test sets
O4. Train a lightweight convolutional neural network baseline model appropriate for smartphone-oriented screening	model checkpoint	Best checkpoint saved, reloadable, and evaluated on held-out test data
O5. Establish an evaluation framework that reports class-specific metrics, confusion matrices, and error analysis outputs for screening tasks	evaluation outputs	Accuracy, precision, recall, F1-score, confusion matrix, and threshold-tuned results reported
O6. Prepare a deployment pathway toward mobile inference through TensorFlow Lite export and associated configuration artifacts	export preparation	Export attempted; deployment configuration documented even though TFLite conversion was unstable in the current environment

1.5 RESEARCH QUESTIONS AND HYPOTHESES

1.5.1 Research Questions

The following research questions guide this dissertation:

RQ1: Which preprocessing and feature representation choices provide a practical trade-off between predictive usefulness and mobile efficiency for cough-based screening?

RQ2: How do label heterogeneity and class imbalance affect screening performance on public cough datasets?

RQ3: Which pipeline design and reporting choices best support reproducibility, credibility, and future clinical validation?

RQ4: Can screening-oriented threshold calibration improve disease sensitivity relative to a default model decision rule?

1.5.2 Working Hypotheses

H1: Log-mel spectrograms paired with a lightweight 2D CNN can provide useful screening performance on public cough datasets while remaining compatible with future mobile deployment.

H2: Explicit label mapping and subject-independent split design improve the credibility and reproducibility of cough-based screening experiments.

H3: Screening-oriented threshold calibration can improve disease sensitivity at an acceptable trade-off in specificity.

1.6 Significance of the Study

This study contributes at both methodological and practical levels. Methodologically, it provides a reproducible pipeline that begins from heterogeneous public datasets and proceeds through metadata auditing, label governance, feature extraction, baseline training, and screening-oriented

evaluation. Practically, it supports the broader goal of making preliminary respiratory screening more accessible through mobile devices.

The work is particularly relevant to low-resource environments where access to clinical infrastructure is limited. A smartphone-based screening system can offer privacy-preserving, offline-capable preliminary assessment and may help direct individuals toward timely follow-up care. Even where final diagnosis remains clinical, an automated screening layer may reduce delay and support triage.

1.7 Scope, Assumptions, and Limitations

1.7.1 Scope

This dissertation draft covers Chapters 1 to 3. It includes the scientific background, literature review, and methodology for dataset acquisition, metadata governance, preprocessing, feature extraction, baseline model development, evaluation design, and deployment preparation.

1.7.2 Assumptions

The study assumes that public cough datasets contain disease-relevant acoustic information sufficient for baseline screening experiments. It also assumes that lightweight models using compact time-frequency features can support later mobile deployment.

1.7.3 Limitations

The present implementation has several limitations. First, tuberculosis data remains controlled-access and was not integrated into the current experimental pipeline. Consequently, the implemented milestone focuses on binary screening of COVID versus healthy/non-target controls, while retaining a broader three-class schema for future work. Second, public datasets may contain weakly verified or self-reported labels, creating unavoidable label uncertainty. Third, the control class is not always strictly healthy; in practice it may include negative or non-target cases. Finally, TensorFlow Lite export was attempted but failed in the current environment due to converter instability, meaning that full deployment remains an engineering step for later completion.

1.8 Organization of the Dissertation

This dissertation is organized into five chapters. Chapter One introduces the study, its motivation, objectives, and scope. Chapter Two reviews the relevant literature on cough acoustics, respiratory sound analysis, machine learning, deployment constraints, and current gaps. Chapter Three presents the materials and methods, including dataset handling, preprocessing, feature extraction, baseline model design, evaluation strategy, and deployment preparation. Chapter Four will present the experimental results and applications. Chapter Five will discuss the findings, draw conclusions, provide recommendations, and suggest directions for future work.

1.9 Expected Outcomes and Practical Gains

The expected outcomes of this work include:

- a cleaned and unified metadata table derived from multiple public datasets,
- a reproducible train/validation/test split design,
- an implemented preprocessing and feature extraction pipeline,
- a lightweight baseline CNN for cough-based screening,
- evaluation artifacts including confusion matrices and class-specific metrics,
- deployment preparation through mobile-oriented export and configuration.

The practical gains include improved access to preliminary respiratory screening, support for privacy-preserving offline operation, and a documented foundation for future clinical validation, tuberculosis extension, and smartphone application development.

1.10 PROJECT TIMELINE

The project evolved in stages consistent with the proposal structure, although some milestones were adapted in response to data availability and implementation constraints.

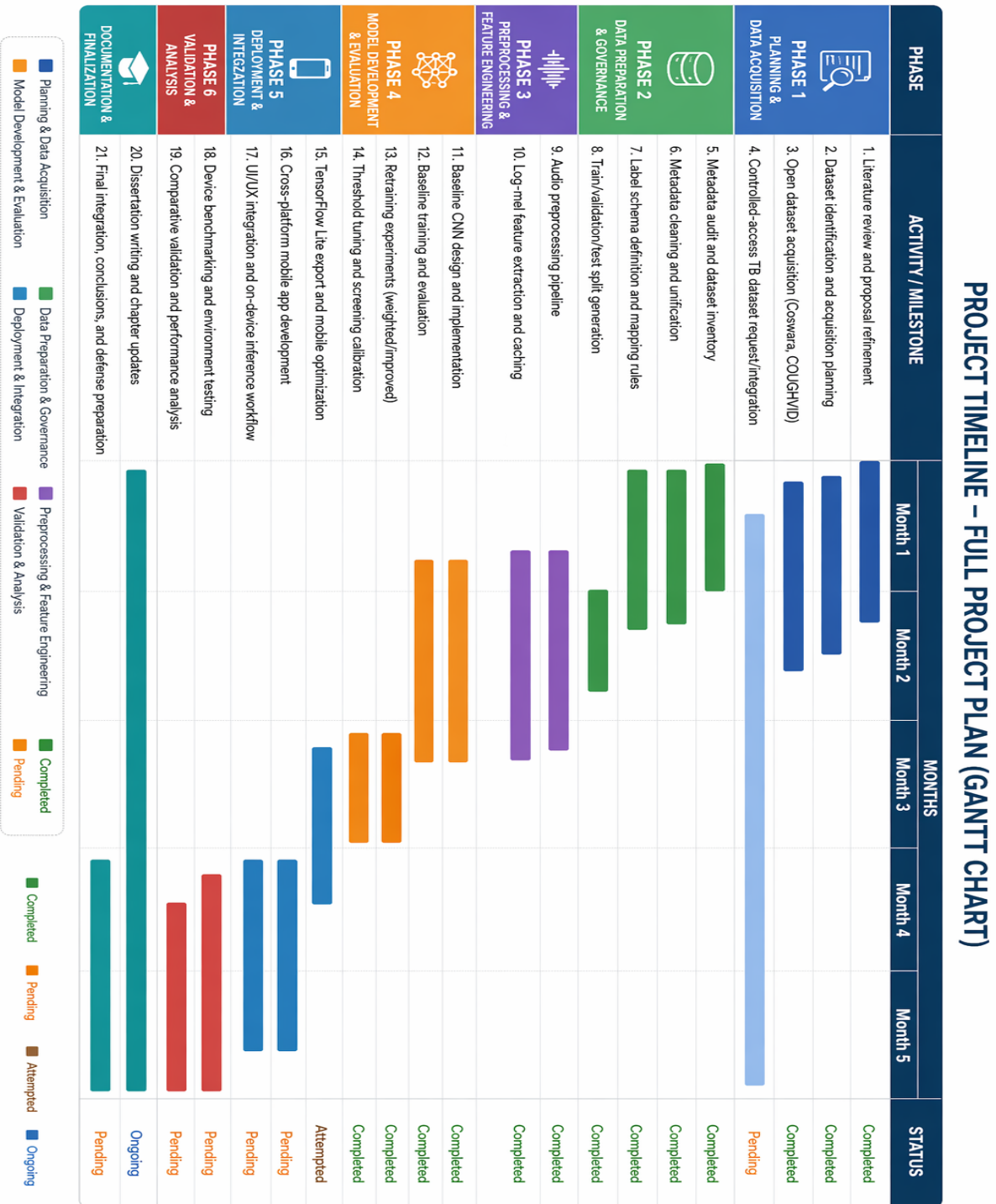


Figure 1.10 Full Project Timeline

At the current point in the work, dataset acquisition, preprocessing, baseline training, threshold-based screening calibration, and artifact generation have been completed for the binary milestone.

CHAPTER TWO: REVIEW OF THE LITERATURE

2.1 Historical Overview of Theory and Research

Respiratory sound analysis has historically been rooted in clinical auscultation, where physicians interpret respiratory sounds and cough characteristics using experience and listening skills. Over time, digital signal processing techniques introduced quantitative analysis of respiratory sounds, enabling the extraction of measurable acoustic features from audio recordings. Early computational work focused on engineered descriptors and conventional classifiers before deep learning became dominant (Pramono et al., 2017; Perna & Tagarelli, 2019).

With the rise of machine learning, researchers began combining handcrafted features with classifiers such as support vector machines and decision trees. These approaches offered initial success but depended strongly on feature engineering choices. The emergence of deep learning shifted the field toward representation learning, allowing neural networks to learn directly from spectrogram-like inputs without requiring all discriminative patterns to be manually specified (Perna & Tagarelli, 2019; Kapetanidis et al., 2024).

The COVID-19 pandemic accelerated this area of research by motivating the collection of large-scale public cough datasets and encouraging work on rapid, remote, and contact-free screening (Brown et al., 2020; Orlandic et al., 2021). As a result, cough-based screening has become a significant applied area within biomedical signal processing and mobile health.

Table 2.1 Comparative Literature Matrix

Study	Data Source / Focus	Main Method	Strength	Limitation	Relevance to This Study
Sharma et al. (2020)	Coswara respiratory sounds for COVID-19	Public dataset creation and benchmark experiments	Provides an important open dataset for cough-based COVID research	Metadata completeness and label heterogeneity vary across records	Directly supports dataset acquisition and label governance
Orlandic et al. (2021)	COUGHVID crowdsourced cough recordings	Large-scale dataset creation for cough analysis	Large cough corpus under real-world recording variability	Crowdsourced labels and variable recording quality introduce noise	Supports dataset scale and motivates preprocessing/quality control
Brown et al. (2020)	Crowdsourced respiratory sounds for COVID-19	ML/DL-based automatic diagnosis	Shows feasibility of crowdsourced respiratory audio classification	Real-world deployment and reproducibility issues remain	Supports proof-of-concept background
Laguarta et al. (2020)	COVID-19 cough recordings	AI-based cough classification	Demonstrates that cough alone can contain disease-relevant information	Reported performance may not generalize across datasets and environments	Supports cough-based screening rationale
Imran et al. (2020)	COVID-19 cough app-based screening	AI pipeline with mobile orientation	Strong relevance to rapid smartphone-based screening	Mobile deployment details and broad real-world validation remain limited	Supports app-oriented motivation
Perna and Tagarelli (2019)	Respiratory anomaly detection	Recurrent neural networks	Early deep learning contribution in respiratory audio	Less deployment-oriented and not cough-focused	Supports historical evolution of deep learning in respiratory analysis

Pramono et al. (2017)	Respiratory sound analysis review	Systematic review	Summarizes classical audio features and signal-processing foundations	Predates the surge in cough-based COVID datasets	Supports preprocessing and feature extraction background
Schuller et al. (2021)	COVID-19 and computer audition overview	Survey / conceptual overview	Broad overview of what sound analysis can contribute during the pandemic	Not a reproducible end-to-end implementation study	Supports framing and deployment motivation
Kapetanidis et al. (2024)	Systematic review of AI for respiratory sound classification	Literature review	Highlights deployment, reproducibility, and robustness gaps	Review study, not an implementation pipeline	Supports critique and gap identification

2.2 Cough Acoustics and Respiratory Disease Signatures

A cough is generated through coordinated respiratory muscle activity, rapid airflow, and airway interaction. The resulting sound may be influenced by inflammation, airway narrowing, mucus presence, tissue changes, and other pathological factors. These influences can affect both temporal and spectral characteristics of the cough signal.

From a signal analysis perspective, respiratory disease signatures in cough sounds may appear through burst structure, duration, spectral emphasis, harmonic content, and time-frequency texture. Such cues may be too subtle or inconsistent for reliable human auditory interpretation, particularly in noisy environments. This motivates the use of machine learning and deep learning approaches capable of learning patterns from many labeled examples.

Because cough is a non-stationary signal, time-frequency representations are especially useful. Spectrograms and mel spectrograms provide a compact representation of how energy changes across time and frequency, making them suitable inputs for convolutional neural networks.

2.3 Public Cough Datasets and Metadata Governance

The growing availability of public cough datasets has made large-scale model development possible. However, the use of these datasets introduces important methodological challenges. Publicly available respiratory sound datasets often differ in label quality, recording format, participant metadata, and health-status definitions. Such heterogeneity can significantly affect downstream training and evaluation (Orlandic et al., 2021; Sharma et al., 2020).

Coswara provides respiratory audio including cough, breathing, and voice samples together with health-related metadata, while COUGHVID provides crowdsourced cough recordings collected under real-world variability (Sharma et al., 2020; Orlandic et al., 2021). These resources are valuable, but they require careful dataset governance. Labels such as healthy, negative, control, recovered, or symptomatic may not always be strictly comparable across datasets. Consequently, any credible pipeline must preserve original labels, define explicit mapping rules, document exclusions, and avoid silent relabeling (Kapetanidis et al., 2024).

Controlled-access tuberculosis datasets present a different challenge. Such datasets may offer stronger label reliability, but access restrictions complicate automation and integration. For this reason, a responsible thesis pipeline should separate open-access and controlled-access data, allowing the absence of the latter to be clearly documented rather than silently ignored.

2.4 Audio Preprocessing and Feature Extraction

Preprocessing is essential in cough-audio analysis because recordings vary widely in duration, amplitude, silence content, and background noise. Typical preprocessing operations include resampling, mono conversion, silence trimming, amplitude normalization, and selection of the most relevant active cough segment (Pramono et al., 2017). In a reproducible pipeline, these operations should be transparent, parameterized, and auditable.

At the same time, preprocessing must be handled carefully. Excessive denoising or aggressive trimming may distort disease-relevant acoustic patterns. Therefore, a strong methodological approach uses quality flags to trace problematic recordings rather than silently deleting them.

Two common feature representations dominate the literature: MFCCs and spectrogram-based features. MFCCs are compact and have long been used in speech and audio analysis. However, log-mel spectrograms are particularly attractive for CNN-based classification because they preserve more time-frequency structure while remaining computationally manageable (Kapetanidis et al., 2024; Hamdi et al., 2023). For mobile-oriented screening, log-mel spectrograms offer a good compromise between informativeness and efficiency.

2.5 Machine Learning and Deep Learning Approaches

Several model families have been applied to respiratory sound and cough analysis. Classical approaches combine engineered features with conventional classifiers, whereas deep learning approaches learn directly from structured feature representations such as spectrograms. CNNs are widely used because they can efficiently learn local time-frequency patterns and remain relatively lightweight when designed with deployment constraints in mind.

Recurrent neural networks and attention-based architectures can capture longer temporal dependencies and may improve performance in some settings. However, these models often increase training and inference complexity. For a mobile-oriented thesis project, a lightweight CNN baseline is a defensible starting point because it is easier to reproduce, easier to export, and more compatible with compact spectrogram inputs.

Transfer learning and self-supervised learning also show promise in health audio analysis, particularly where labels are limited or noisy. Nevertheless, such methods may introduce dependencies on larger pretrained models and complicate mobile deployment. As a result, they are better positioned as later extensions rather than first-stage baseline methods.

2.6 Evaluation Protocols and Screening Metrics

Reliable evaluation requires more than a single accuracy value. In cough-based screening, train/validation/test splits should be carefully designed to reduce subject leakage, especially when multiple recordings may come from the same participant. Subject-independent splitting therefore provides stronger credibility than random sample-level partitioning when subject identifiers are available (Kapetanidis et al., 2024).

For screening tasks, class-specific recall is especially important. A model that achieves high overall accuracy by always predicting the majority class may still be clinically unhelpful if it misses most positive cases. Confusion matrices, precision, recall, and F1-scores therefore provide a more informative picture than accuracy alone. Threshold tuning may also be necessary, because screening often prioritizes sensitivity over specificity (Schuller et al., 2021; Kapetanidis et al., 2024).

2.7 Mobile/Edge Deployment and Model Optimization

Mobile deployment introduces additional constraints beyond model accuracy. A deployable system must consider model size, inference latency, preprocessing cost, battery usage, privacy, and offline operation. TensorFlow Lite is a common target for edge deployment because it supports compact inference on consumer devices (TensorFlow Lite Documentation, n.d.). However, successful mobile deployment depends not only on training performance but also on stable conversion and compatibility of the software environment.

The literature shows that many works mention mobile deployment in principle but do not provide reproducible export steps, conversion artifacts, or deployment troubleshooting (Imran et al., 2020; Kapetanidis et al., 2024). This dissertation addresses that gap by explicitly incorporating an export stage into the pipeline and documenting both success criteria and converter limitations.

2.8 Critique of Current Literature and Identified Gaps

Despite promising results in cough-based disease screening, several recurring weaknesses remain in the literature. First, label governance is often poorly described, especially where mixed-control groups or self-reported labels are used. Second, reproducibility is limited when authors do not release cleaned metadata tables, split files, or scripts. Third, evaluation protocols may not always guard sufficiently against subject leakage or over-optimistic reporting. Fourth, deployment claims are frequently not accompanied by real export artifacts or operational mobile constraints.

These gaps are important because cough screening is especially vulnerable to misleading results. A model may appear highly accurate while remaining insensitive to the disease class. Similarly, a model may perform well in controlled experiments but fail to generalize across devices and environments.

2.9 Contribution of This Study

This dissertation contributes by focusing not only on model training but also on the reproducible engineering pipeline that supports credible screening research. Specifically, it contributes:

- explicit dataset auditing and metadata standardization,
- a documented label schema with traceable mapping rules,
- subject-aware splitting procedures,
- reproducible feature extraction using fixed-size log-mel spectrograms,
- a lightweight CNN baseline aligned with mobile constraints,
- a screening-oriented threshold calibration strategy,
- and a deployment pathway that includes export preparation and documented converter limitations.

CHAPTER THREE: MATERIALS AND METHODS

3.1 Research Design and Methodology

This study follows an applied machine learning methodology with a deployment-oriented perspective. Rather than treating classification as an isolated modeling task, the study is structured as a complete and reproducible pipeline that transforms raw cough datasets into trainable feature tensors, a baseline model, evaluation artifacts, and deployment preparation outputs. The research design is iterative: a baseline pipeline is implemented, evaluated, audited for failure modes, and refined.

The central methodological principle is traceability. Raw labels are preserved, mapping rules are made explicit, and intermediate artifacts are saved to support replication and future extension. This approach is particularly important in public cough datasets, where labels, metadata, and recording quality can vary substantially across sources.

3.2 System Overview and Workflow

The implemented system is organized into the repository structure `thesis-cough-screening/` and follows the workflow below:

1. Download open datasets and generate manual instructions for controlled-access data.
2. Audit metadata and inventory files.
3. Clean per-dataset metadata and build a unified metadata table.
4. Create train/validation/test splits with subject-independent logic where possible.
5. Preprocess audio and compute fixed-size log-mel spectrogram features.
6. Train a lightweight CNN baseline model.
7. Evaluate the baseline and compare threshold-calibrated screening behavior.
8. Attempt mobile-oriented export through TensorFlow Lite.

This workflow was executed in practice for the current binary milestone.

Table 3.2 Label Schema and Mapping Rules

Field / Label Level	Description	Example / Rule	Purpose
label_raw	Original dataset-provided label	e.g., positive, healthy, recovered, negative	Preserves source traceability
label_multiclass	Standardized thesis label	TB, COVID, HEALTHY_OR_NONTARGET	Creates a controlled common schema across datasets
label_binary	Binary screening label for current milestone	COVID vs HEALTHY_OR_NONTARGET	Supports implemented 2-class model
include_for_training	Boolean flag indicating if sample is trainable	Yes/No	Prevents silent inclusion of unmappable samples
notes	Explanation of mapping decision or exclusion	e.g., weak label, mixed control, missing metadata	Improves transparency and auditability

3.3 Data Sources and Acquisition

3.3.1 Open Datasets

The present implementation uses two public cough-audio sources:

Coswara — a public respiratory audio dataset containing cough, breathing, and voice recordings together with participant metadata. For this study, the cough subset was used. The acquired and extracted data contained **24,716 audio files** corresponding to **2,746 subjects**.

COUGHVID — a public crowdsourced cough dataset containing cough recordings and associated metadata. The acquired dataset contained **27,550 files**, from which healthy/COVID-relevant records were retained for unified metadata construction.

3.3.2 Controlled-Access Dataset

Tuberculosis cough data was planned as part of the broader thesis scope. However, because the relevant TB dataset required controlled access, it was not integrated into the current implemented training milestone. The pipeline was designed to accommodate such data later through documented manual acquisition steps rather than simulated automation.

Table 3.3 Default Preprocessing and Feature Parameters

Parameter	Default Value	Description / Purpose
Target sample rate	16,000 Hz	Standardizes audio for training and later deployment
Channel mode	Mono	Reduces variability and compute cost
Silence trimming	Enabled	Removes long silent regions
Loudest-segment extraction	Enabled	Focuses on the most cough-relevant region
Peak normalization	Enabled	Stabilizes amplitude scale across recordings
Noise reduction	Optional (Wiener filtering)	Available in pipeline but not emphasized in baseline
Quality flags	Enabled	Tracks too-short, too-long, clipping, and low-quality samples
Feature type	Log-mel spectrogram	Primary input representation for CNN
Mel bins	64	Compact time-frequency resolution
FFT size	1024	Spectral analysis window size
Hop length	256	Time step between frames
Window length	512	STFT window size
Frequency range	50–8000 Hz	Focuses on cough-relevant spectral region
Maximum frames	256	Produces fixed-size input tensors
Output tensor shape	(64, 256, 1)	CNN-compatible single-channel input

3.4 Label Schema and Data Governance

The intended label schema for the full thesis is a three-class structure:

- **TB**
- **COVID**
- **HEALTHY_OR_NONTARGET**

However, because TB data was not yet available in the implemented pipeline, the current milestone was executed as a binary screening task between **COVID** and **HEALTHY_OR_NONTARGET**.

The control category is intentionally named **HEALTHY_OR_NONTARGET** rather than simply healthy because public datasets frequently include negative or non-target cases that may not be strictly asymptomatic healthy individuals. This naming choice improves scientific transparency.

For each dataset, the pipeline retains the original `label_raw` field and derives mapped labels through explicit rules rather than ad hoc manual edits. The metadata pipeline also retains explanatory notes and inclusion flags to document why samples are included or excluded from training.

A standard unified schema was used to combine heterogeneous datasets. Key fields include sample identifiers, subject identifiers, file paths, recording type, raw labels, mapped labels, audio properties, quality flags, split assignment, and notes.

The implemented unified metadata file contained **18,649 samples** after cleaning and label mapping.

Table 3.4 Training Configuration and Hyperparameters

Item	Value / Setting	Notes
Model type	Lightweight 2D CNN	Baseline architecture for binary screening
Input representation	Log-mel spectrograms	Fixed-size tensors
Task	Binary classification	COVID vs HEALTHY_OR_NONTARGET
Loss function	Sparse categorical cross-entropy	Used for current training setup
Optimizer	Adam	Baseline optimizer

Learning rate	1e-3	Default baseline configuration
Batch size	16	Configurable
Epochs	15 (baseline setting)	Early stopping/checkpointing used to prevent overfitting
Validation monitor	val_loss	Used for checkpointing/early stopping
Split strategy	Subject-independent where possible	Reduces leakage
Regularization	Dropout + batch normalization	Included in architecture
Class imbalance handling	Explored via threshold tuning and retraining experiments	Threshold tuning proved more useful than class-weighted retraining in the current milestone

3.5 Split Strategy

Train, validation, and test splits were generated from the unified metadata table. Where subject identifiers were available, the pipeline used subject-independent splitting in order to reduce the risk of data leakage. This ensures that recordings from the same participant do not appear in both training and testing stages.

The final split sizes for the current milestone were:

- **Training set:** 17,141 samples
- **Validation set:** 754 samples
- **Test set:** 754 samples

Split files were saved as CSV artifacts for reproducibility and later auditing.

Table 3.5 Evaluation Metrics and Interpretation for Screening

Metric	Meaning	Screening Interpretation
Accuracy	Overall proportion of correct predictions	Useful but insufficient on its own in imbalanced screening tasks
Precision (disease class)	Proportion of predicted disease cases that are truly positive	Reflects reliability of positive alerts
Recall / Sensitivity (disease class)	Proportion of actual disease cases correctly detected	Critical for minimizing missed high-risk cases
Recall / Specificity-like control recovery	Proportion of control cases correctly identified	Important to avoid excessive false alarms
F1-score	Harmonic mean of precision and recall	Useful summary of disease-class detection balance
Confusion matrix	Counts of TP, FP, FN, TN	Reveals concrete failure modes
Threshold-calibrated comparison	Performance at different decision thresholds	Supports screening-oriented trade-off selection

3.6 Preprocessing Procedures

Audio preprocessing was applied to standardize recordings while preserving relevant cough information. The implemented procedures included:

- resampling to **16 kHz**,
- conversion to mono,
- silence trimming,
- loudest-segment extraction to isolate the main cough event,
- peak normalization,
- generation of quality flags for unusual recordings.

Rather than silently dropping problematic recordings, the system tracked issues such as very short duration or low-quality segments through flags, allowing later inspection and sensitivity analysis.

Optional Wiener filtering was included as a noise-reduction component in the codebase but was not emphasized in the baseline to avoid introducing uncontrolled distortions.

Table 3.6 Preliminary Implementation Status and Achieved Milestones

Stage	Planned Task	Current Status	Evidence / Output
Dataset acquisition	Acquire public cough datasets	Completed for Coswara and COUGHVID	Coswara extracted; COUGHVID acquired
TB integration	Add TB data into training pipeline	Pending	Controlled-access dataset not yet integrated
Metadata audit	Inspect dataset structure and metadata	Completed	Audit outputs generated
Metadata unification	Build common metadata schema	Completed	Unified metadata file created
Split generation	Create reproducible train/val/test splits	Completed	Subject-independent split CSVs saved
Preprocessing	Standardize audio and extract cough segments	Completed	Log-mel feature pipeline executed
Feature extraction	Generate cached training tensors	Completed	.npy feature arrays saved
Baseline training	Train lightweight CNN	Completed	Baseline checkpoint saved
Retraining experiments	Explore weighting / alternative training	Completed but not selected	Weighted/improved variants underperformed
Threshold calibration	Tune decision threshold for screening	Completed	Threshold 0.35 selected
Final current screening result	Evaluate on held-out test set	Completed	Best result: 67.24% accuracy, 55.39% COVID recall
TFLite export	Export model for mobile deployment	Attempted but unstable	Converter failed in current environment
Mobile app integration	Integrate into smartphone app	Pending	Reserved for later chapter/stage

3.7 Feature Engineering: Log-Mel Spectrograms

The principal feature representation used in this study is the **log-mel spectrogram**. This representation captures the time-varying spectral energy of cough signals in a compressed form that is suitable for image-like processing by convolutional neural networks.

The feature extraction pipeline computes an STFT-based spectrogram, maps it through a mel filterbank, and applies a logarithmic compression to reduce dynamic range. This produces a compact time-frequency representation that preserves local acoustic structure.

The default implementation parameters were:

- **mel bins:** 64
- **FFT size:** 1024
- **hop length:** 256
- **window length:** 512
- **frequency range:** 50–8000 Hz
- **maximum frames:** 256

Each feature tensor was padded or truncated to a fixed frame length and normalized to a stable numeric range. This produced consistent inputs of shape suitable for training and later deployment.

Feature arrays were cached into .npy files for the train, validation, and test sets in order to reduce repeated computation and ensure reproducibility across training runs.

3.8 Model Architecture and Training Protocol

The baseline classifier implemented in this study is a lightweight 2D CNN designed with mobile efficiency in mind. The model processes log-mel spectrogram tensors and outputs class probabilities for the binary screening task.

The architecture consists of:

- stacked convolutional blocks,

- batch normalization layers,
- pooling operations,
- global average pooling,
- dropout regularization,
- a dense hidden layer,
- and a final softmax output layer.

This lightweight structure was selected as a practical baseline because it is easier to debug, easier to reproduce, and more compatible with later export and on-device inference than heavier architectures.

Training used sparse categorical cross-entropy and the Adam optimizer. Model checkpointing and validation monitoring were used to preserve the best-performing baseline. During implementation, binary label remapping was introduced to align the current two-class milestone with the broader original schema.

Additional retraining experiments using weighting and modified procedures were also attempted, but the baseline model remained the most useful model for disease detection in the current stage.

3.9 Testing and Evaluation Plan

The evaluation framework in this study was designed for screening rather than strict diagnosis. Accordingly, it emphasizes not only overall accuracy but also disease-class recall, confusion matrices, and threshold-sensitive behavior.

The evaluation outputs include:

- accuracy, precision, recall, and F1-score,
- confusion matrix,
- threshold-calibrated screening comparisons,
- and error-analysis artifacts for later qualitative inspection.

For screening tasks, disease recall is especially important because false negatives may delay follow-up care. This motivated the use of threshold calibration in addition to the default model decision rule.

3.9.1 Pilot/Implemented Baseline Status

The current implemented milestone produced a complete baseline training and evaluation cycle. A default-threshold baseline model achieved stronger overall accuracy but low COVID sensitivity. A screening-oriented threshold calibration was therefore applied on validation data. The selected threshold of **0.35** was then evaluated on the held-out test set.

This threshold-tuned operating mode produced the best current screening result, with improved COVID recall relative to the model's default decision rule. This result is treated as a preliminary but meaningful milestone supporting later expansion and benchmarking.

3.10 Instrumentation

3.10.1 Software Environment

The implementation used Python-based scientific and machine learning libraries, including TensorFlow, librosa, NumPy, pandas, SciPy, scikit-learn, matplotlib, and soundfile. The pipeline was organized into modular source directories for dataset handling, preprocessing, feature extraction, modeling, evaluation, and export.

3.10.2 Compute Environment

The implementation was executed in a cloud-based environment for scalable data processing and training. The working directory was maintained under `/workspace/lively`. This environment supported dataset extraction, feature generation, model training, and evaluation. However, TensorFlow GPU support and TensorFlow Lite conversion stability were limited in the current container configuration, causing some deployment-related operations to fall back to CPU or fail during export.

3.10.3 Reproducibility Artifacts

Intermediate and final artifacts were stored under project directories such as `metadata/`, `data/processed/`, `models/`, and `reports/`. This included unified metadata tables, split CSVs, feature arrays, model checkpoints, and summary reports.

3.11 Ethical Considerations

This dissertation uses secondary datasets and does not involve direct participant recruitment within Chapters 1 to 3. Ethical considerations therefore focus on responsible use of public and controlled-access datasets, appropriate acknowledgment of dataset provenance, and accurate framing of the model as a screening aid rather than a diagnostic system.

The study also adopts a privacy-conscious orientation by targeting eventual on-device inference and minimizing dependence on server-side processing. Future clinical validation, participant recruitment, or primary data collection will require appropriate institutional ethical approval and informed consent.

3.12 Scope of Current Research

Up to Chapter 3, this dissertation provides:

- the motivation and literature foundation for cough-based respiratory screening,
- a documented methodology for dataset governance and metadata standardization,
- preprocessing and feature engineering using log-mel spectrograms,
- a lightweight baseline CNN model for binary screening,
- evaluation methodology emphasizing screening-oriented calibration,
- and deployment preparation through attempted TFLite conversion and configuration artifacts.

The current implemented work stops before full multi-class respiratory integration, full mobile user interface development, and clinical validation. Tuberculosis integration, stable export conversion, and complete application deployment remain later stages of the broader thesis.

3.13 Reproducibility Procedure

The implemented workflow is executed in the project repository in the following order:

1. dataset download and manual controlled-access setup,
2. metadata audit and unification,
3. split creation,
4. preprocessing and feature caching,
5. baseline model training,
6. evaluation and threshold selection,
7. deployment preparation and export attempt.

This stage-wise artifact generation ensures that intermediate results can be inspected, reproduced, and extended in later work. The methodology therefore supports both academic transparency and practical engineering continuity.

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